Interactive Gantt Chart Builder

by Fraser Davies

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by Dr Pierre Le Bras

Heriot-Watt University
School of Mathematical and Computer Sciences

18th November 2025

Authorship Declaration

Course code and name	F20PA - Honours Project
Type of assessment	Individual
Coursework title	Interactive Gantt Chart Builder
Student name	Fraser Davies
Student ID	H00403114

By signing this form:

- I declare that the work I have submitted for individual assessment or the work I have contributed to a group assessment is entirely my own. I have not taken the ideas, writings or inventions of another person and used them as if they were my own. My submission or my contribution to a group submission is expressed in my own words. Any uses made within this work of the ideas, writings or inventions of others, or of any existing sources of information (books, journals, websites, etc.) are properly acknowledged and listed in the references and/or acknowledgements sections.

- I confirm that I have read, understood and followed the University's Regulations on academic misconduct as published on the University's website, and that I am aware of the penalties that I will face should I not adhere to the University Regulations.

- I confirm that I have read, understood and avoided the different types of plagiarism explained in the University guidance on Academic Integrity and Plagiarism.

Student signature: Fraser Davies

Date: 18th November 2025

Your work will not be marked if a signed copy of this form is not included with your submission.

Acknowledgements

Acknowledgements text.

Abstract

Abstract text.

Table of Contents

1 Introduction

1.1 Background and Motivation

Project management is critical for students in higher education, particularly those undertaking substantial individual projects such as honours dissertations. Dissertations present unique challenges as they are often the largest and most complex projects students have encountered, requiring sustained effort over extended periods while juggling multiple responsibilities [26]. Effective planning, scheduling, and progress monitoring are essential skills that students must develop to successfully navigate these complex, long-term academic endeavours. However, many students struggle with translating abstract project requirements into concrete, manageable tasks with realistic timelines and dependencies, finding themselves overwhelmed by the scope and duration of dissertation work [26].

Gantt charts have emerged as one of the most widely adopted project management tools, originally developed by Henry Gantt in the early 20th century to visualise project schedules through horizontal bar representations [9, 28]. These visual tools effectively display task durations, dependencies, and timelines, making them invaluable for planning and tracking project progress [7]. In educational contexts, Gantt charts serve as pedagogical tools that help students manage their increasing responsibilities while developing time management skills. Their graphical representation visualises beginning dates, ending dates, and duration parameters, helping to track various projects over specific time periods [3].

Despite their proven utility in professional project management, traditional Gantt chart tools often present usability challenges for students who lack prior project management experience. Many existing solutions are designed for team-based projects and enterprise environments, making them overly complex for individual academic projects. Furthermore, the widespread adoption of visualisation tools in educational settings has been hindered by usability problems and the time required for instructors and students to learn and incorporate these tools into their workflows.

At Heriot-Watt University (HWU), honours project students are currently required to submit project management plans as part of their dissertation deliverables. However, the existing project management system lacks integrated tools that guide students through the process of creating well-structured Gantt charts with appropriate task hierarchies, dependencies, and milestones. This gap in the current system motivated the development of an interactive Gantt chart builder specifically tailored to the needs of honours project students and their supervisors.

1.2 Aim and Objectives

The primary aim of this project is to design, develop, and evaluate an interactive Gantt chart builder that will be integrated into the existing HWU honours project management system.

This tool will provide structured guidance to help students create comprehensive project plans while offering supervisors visibility into student progress and planning quality.

The specific objectives of this project are:

(1) To conduct comprehensive background research on Gantt chart design principles, project management methodologies, and interactive visualisation techniques relevant to educational contexts
(2) To gather and analyse requirements from key stakeholders, including current system developers, supervisors, and students, to ensure the tool meets the needs of its target users
(3) To design an intuitive user interface that guides students through the process of defining milestones, tasks, task hierarchies, and dependencies without requiring prior project management expertise
(4) To develop an interactive Gantt chart visualisation using D3.js (Data-Driven Documents), a powerful JavaScript library that enables the creation of dynamic and interactive data visualisations in web browsers using HTML, SVG, and CSS [8, 30], ensuring seamless integration with the existing HWU project system's technological stack
(5) To implement functionality that allows students and supervisors to explore project timelines, track progress, and reevaluate goals as projects evolve
(6) To provide export capabilities that enable students to include their Gantt charts and associated data in their project documentation
(7) To conduct a rigorous usability evaluation through user studies with students and supervisors to assess the effectiveness of the tool and its guidance features

1.3 Organisation

The organisation of this research report is as follows:

- Section 2: Presents the background research, discussing related work found in the technical literature and its relevance to the project.
- Section 3: ...
- Section 4: ...

2 Background Research

2.1 Gantt Charts and Project Management

The Gantt chart traces its origins to the early twentieth century and the scientific management movement. American mechanical engineer and management consultant Henry Gantt first described a version of his charts in 1903, published alongside Frederick W. Taylor's influential work on shop management, though scholars debate whether their widespread adoption began during the First World War [33]. Conceived as production planning tools, Gantt charts monitored worker performance and machine utilisation, embodying the principles of scientific management through rational analysis, clear task durations, and systematic improvement [33]. Crucially, they addressed the factory-wide machine loading problem, without which Taylor's planning office would have been unworkable [33]. While project management applications of Gantt charts were initially limited, their paper-based forms declined in the 1950s and 1960s. They regained popularity with computerised methods in the mid-1960s and were further revitalised by microcomputer project software in the 1980s. Today, Gantt charts have evolved into flexible, software-based tools central to modern project management [33].

Building upon Henry Gantt's original design, modern Gantt charts have evolved to incorporate sophisticated features that support complex project planning. A comprehensive Gantt chart integrates multiple elements: [7, 28]

- A vertical list of tasks or activities on the left side.
- A horizontal timeline indicating project duration (days, weeks, or months)
- Graphical bars representing individual task durations and their temporal placement.
- Lines or arrows indicating dependencies between tasks.
- Labels showing team members or assignees.
- Special markers (typically diamonds) that denote milestones or important project achievements.

Task dependencies are fundamental to effective Gantt chart usage, defining the relationships between tasks and dictating the order in which activities must occur. Project management identifies four primary dependency types [21, 27]. The Finish-to-Start (FS) dependency, where a successor task cannot commence until its predecessor concludes, represents the most common relationship type [19]. Start-to-Start (SS) dependencies allow a task to begin only after another task has also began, enabling parallel workflows that share a start point. Finish-to-Finish (FF) relationships constrain task completion, requiring one task to finish before another can conclude. Finally, the Start-to-Finish (SF) dependency, the least common type, denotes that a task cannot finish until another task begins, typically appearing in handover or transition scenarios.

Beyond dependencies, Gantt charts visualise the critical path the longest sequence of dependent tasks that determines the project's minimum completion time. This enables project managers to distinguish between tasks that are critical to meeting deadlines and those that possess scheduling flexibility [2]. This distinction proves particularly valuable for resource allocation and risk management decisions.

Gantt charts occupy a distinct position within the broader landscape of project management methodologies. Unlike Kanban boards, which emphasise workflow states and work-in-progress limits while focusing on the volume of work completed, in progress, and not yet started, Gantt charts provide explicit temporal representations of task scheduling and dependencies displayed on a visual timeline [23]. Similarly, while Program Evaluation and Review Technique (PERT) charts focus on identifying critical paths through network diagrams using nodes and arrows to represent task dependencies, Gantt charts offer a more intuitive timeline-based bar chart visualisation that facilitates stakeholder communication and progress tracking [14, 22].

Within project management practice, Gantt charts serve multiple critical functions that extend beyond simple schedule visualisation. As planning tools, they enable project managers to structure complex projects by breaking them into manageable tasks with defined durations and sequences [31]. Their visual representation of timelines, task dependencies, and resource allocation facilitates effective coordination and communication among project stakeholders [31], ensuring all team members maintain a shared understanding of project progress and responsibilities [31]. Furthermore, Gantt charts function as monitoring instruments through features such as baseline comparisons and progress tracking, allowing project managers to identify deviations early and make informed decisions about resource reallocation or timeline adjustments [31]. This combination of planning, communication, and monitoring capabilities has established Gantt charts as one of the most frequently used project management tools, ranking fourth among 70 tools and techniques in a survey of 750 project managers [4, 31].

The application of Gantt charts and project management tools in educational contexts, particularly for individual student projects, remains an underexplored area in academic literature. While Bednjanec and Tretinjak demonstrate educators' use of Gantt charts for planning school activities and curriculum development [3], research on project management tools in higher education focuses predominantly on institutional IT departments [12] and collaborative group work [10]. This focus on institutional and collaborative applications leaves a gap in understanding how individual students might effectively use Gantt charts for dissertation management. Existing project management research centres on professional contexts [4], where tools are designed for team-based environments with different requirements than individual academic projects.

2.2 Gantt Charts in Educational Contexts

Within educational contexts, Gantt charts serve multiple pedagogical functions. For educators, Gantt charts provide a structured approach to planning school activities, curriculum development, and lesson planning, with their graphical representation of beginning dates, ending dates, and duration parameters proving particularly effective for tracking various educational activities over specific time periods [3]. Beyond their use by educators, Gantt charts offer valuable support for students undertaking complex academic projects such as dissertations. The ability to visualise project timelines and decompose large, complex undertakings into manageable components with clear deadlines proves particularly valuable when managing extended research projects [26]. This visual representation of project structure enables both students and educators to conceptualise and monitor multiple activities across extended timeframes [3].

However, students often encounter challenges when applying Gantt charts to individual academic projects, particularly concerning the maintenance and updating of project timelines throughout extended research periods. Research with PhD students reveals that while Gantt charts can be beneficial for planning, the unpredictable nature of academic research makes it difficult to maintain detailed charts over multi-year projects, with students reporting frustration when plans deviate from initial timelines [1]. This psychological barrier to updating charts is compounded by the investment required: as one PhD student observes, "if one invest a lot of time and energy in creating a detailed Gantt chart, then one is not gonna be willing to adjust this chart when things didn't work out as planned" [1]. This reluctance to revise detailed plans creates a tension between the effort invested in initial planning and the flexibility required for successful academic project management.

Traditional Gantt chart tools required frequent manual updates to remain accurate, which created a significant maintenance burden. Although modern software reduces this through automation, regular updates are still necessary, and increasing project complexity, particularly in large projects like a dissertation, makes Gantt charts challenging to manage [18]. Modern interactive Gantt chart platforms address many of these challenges through features such as drag-and-drop task scheduling, automatic dependency adjustments, and real-time collaborative updates, which significantly reduce the effort required to maintain project schedules and enable more flexible adaptation to changing project requirements [29]. These interactive capabilities prove particularly valuable for complex projects where tasks and timelines must be frequently adjusted, as they allow users to modify schedules efficiently without the cumbersome manual recalculation required by static chart implementations [29].

The distinction between individual and collaborative project contexts significantly influences the application of project management tools like Gantt charts. In team-based academic projects, collaborative project management platforms provide mechanisms for distributed task assignment, shared responsibility, and collective accountability, enabling group members to coordinate schedules, track individual contributions, and maintain visibility of project progress [10]. These collaborative features address common challenges in group work, including coordination costs, workload equity concerns, and the need for clear role definition among team members [10]. However, individual academic projects such as dissertations present fundamentally different challenges. Unlike team projects where collective objectives and shared evaluation create mutual accountability, dissertation students bear sole responsibility for project completion and are evaluated individually on their work [24]. Research into academic project management reveals that while research involving multiple students may appear collaborative, the structure actually consists of professor-student duos where each student manages their own sub-project as a junior partner rather than as part of an integrated team [24]. This absence of collective objectives and team-based evaluation means individual dissertation students must rely entirely on autonomous use of project management tools without the distributed workload or accountability structures present in genuine team contexts [4]. The challenges become especially pronounced when individual students encounter the psychological barriers associated with maintaining detailed project schedules throughout extended research periods, as the sole responsibility for both planning and plan maintenance creates tension between the effort invested in initial timeline development and the flexibility required for adaptive project management [1].

2.3 Interactive Data Visualisation

Effective data visualisation relies on fundamental principles that ensure clarity and comprehension. Key design principles include knowing the audience, maintaining simplicity, choosing appropriate chart types, using colour strategically whilst maintaining accessibility, and providing proper context through titles, labels and scales [13]. Interactive visualisation differs fundamentally from static visualisation in its approach to user engagement: whilst static visualisation employs an author-driven approach where a single data story is communicated, interactive visualisation adopts a reader-driven approach that enables users to extract multiple views and explore data stories dynamically through inputs such as mouse clicks or keyboard commands [17]. Interactive graphics can provide crucial information that cannot be obtained from static graphics, making them valuable tools for promoting transparency and enabling additional exploration of datasets [32]. The benefits of interactivity in educational contexts are substantial, as:

- Interactive visualisations transform data into engaging learning experiences that enhance comprehension, improve retention, and enable personalised exploration of complex concepts [11].
- User interaction paradigms for visualisation tools encompass several distinct approaches: drag-and-drop interactions enable intuitive grouping, reordering and moving of objects through direct manipulation, providing clear signifiers and immediate visual feedback throughout the interaction [15].
- Form-based interfaces present users with structured options through menus and input fields that guide interaction, allowing menu selection by keying in associated numbers or letters whilst graphical menus employ pointing devices [6].
- Direct manipulation interfaces allow users to act on displayed objects using physical, incremental actions with immediate visual feedback—exemplified by resizing graphical elements or repositioning items by dragging [25].

These interaction paradigms collectively support varied user needs and task requirements, with direct manipulation particularly valued for its ability to exploit perceptual and motor skills whilst reducing reliance on linguistic comprehension [25].

3 Methodology

3.1 Stakeholders and Use Cases

Students Supervisors Project system developers

3.2 Requirement Analysis

The requirement analysis for the Interactive Gantt Chart Builder was conducted through examination of existing literature on project management tools in educational contexts, consultation of stakeholder needs identified through preliminary discussions with the current system developers, and analysis of the challenges students face when managing individual dissertation projects. This process identified five core functional requirements that form the foundation of the proposed system.

The requirements are categorised using the MoSCoW method into Must have (M), Should have (S), Could have (C), and Won't have (W) [20]. To label them they are indicated FR[#] to represent Functional Requirement number [#] and NFR[#] to represent Non-Functional Requirement number [#]. Each requirement has a User Story[5] format description to clarify the requirements purpose and context.

- **FR1 - M - Project Data Creation and Management**
  As a student, I must be able to create and store my project information (title, description, start date, end date) so that I have a foundation for my Gantt chart.
- **FR2 - M - Task Creation and Editing**
  As a student, I must be able to create, edit, and delete tasks with names, descriptions, start dates, and end dates so that I can break down my dissertation into manageable components.
- **FR3 - M - Task Dependencies**
  As a student, I must be able to define dependencies between tasks (Finish-to-Start relationships) so that my project plan reflects the logical sequence of work.
- **FR4 - M - Gantt Chart Visualisation**
  As a student, I must be able to view my tasks and milestones as a visual Gantt chart using D3.js so that I can understand my project timeline at a glance.
- **FR5 - M - Data Saving**
  As a student, I must be able to save my project data to a database (SQLite) so that my work is not lost between sessions.
- **FR6 - M - Milestone Creation**
  As a student, I must be able to create milestones with specific dates so that I can mark key deliverables and deadlines in my dissertation timeline.
- **FR7 - S - Guided Task Creation Interface**
  As a student, I should receive contextual help and validation when creating tasks so that I create well-structured, realistic project plans.

- **FR8 - S - Interactive Timeline Adjustment**
  As a student, I should be able to drag and drop tasks on the Gantt chart to adjust dates and durations so that I can quickly adapt my plan as circumstances change.
- **FR9 - S - Automatic Dependency Adjustment**
  As a student, when I modify a task's dates, the system should automatically update dependent tasks so that I maintain a logically consistent schedule.
- **FR10 - S - Progress Tracking and Status Updates**
  As a student, I should be able to mark tasks as not started, in progress, or completed so that I can track my actual progress against my plan.
- **FR11 - S - Progress Indicator and Timeline Comparison**
  As a student, the system should tell me if I am ahead of schedule, on track, or behind schedule so that I can take corrective action when needed.
- **FR12 - S - Gantt Chart Export (PNG/JPEG)**
  As a student, I should be able to export my Gantt chart as an image (PNG/JPEG) so that I can include it in my dissertation appendix and supervision meeting documents.
- **FR13 - S - Supervisor View-Only Access**
  As a supervisor, I should be able to view my students' Gantt charts in read-only mode so that I can monitor their progress and provide guidance.
- **FR14 - S - Task Hierarchy**
  As a student, I should be able to create subtasks under parent tasks so that I can organise complex phases of work hierarchically.
- **FR15 - S - Impossible Dependencies**
  As a student, the system should prevent me from creating logically impossible schedules.
- **FR16 - S - Timeline Zoom and Pan Controls**
  As a student, I should be able to zoom in/out and pan across the timeline so that I can view different levels of detail from my entire project to individual weeks.
- **FR17 - S - Hover Tooltips with Task Details**
  As a student, when I hover over a task bar, I should see detailed information (task name, dates, duration, dependencies) so that I can quickly access task details without navigating away.
- **FR18 - C - Kanban Board View**
  As a student, I could have an alternative Kanban board view of my tasks so that I can visualise my work in a different format that shows current work in progress.
- **FR19 - C - Task Categories/Tags**
  As a student, I could assign categories or tags to tasks (e.g., "Research", "Writing", "Data Collection") so that I can filter and organise my work by type.
- **FR20 - C - Multiple Dependency Types**
  As a student, I could define different dependency types (Start-to-Start, Finish-to-Finish, Start-to-Finish) beyond the basic Finish-to-Start so that I can model more complex task relationships.
- **FR21 - C - Critical Path Highlighting [4]**

As a student, I could view the critical path highlighted on my Gantt chart so that I can identify which tasks have no scheduling flexibility [4].
- **FR22 - C - Supervisor Comment and Feedback System**
As a supervisor, I could add comments and feedback on specific tasks or the overall project plan so that I can provide guidance directly within the tool.
- **NFR2 - M - Web Browser Compatibility** As a student, I must be able to access and use the Gantt chart builder on any modern web browser (Chrome, Firefox, Safari, Edge) without installing additional plugins or software so that I can work on my project plan from any computer.
- **NFR3 - M - Data Security and GDPR Compliance** As a student, the system must store all my project data securely and be GDPR-compliant.
- **NFR4 - S - Response Time Performance** As a student, the system should render my Gantt chart and respond to my interactions within 2 seconds (for projects with up to 100 tasks) so that I can work efficiently without frustrating delays that discourage regular use.
- **NFR5 - S - Intuitive User Interface** As a student, the system should provide a clear, intuitive interface that I can use effectively without extensive training so that I can focus on planning my dissertation rather than learning how to use the tool.
- **NFR6 - S - Mobile-Friendly Responsive Design** As a student, the system should adapt its interface for my tablet and mobile devices so that I can view and ideally edit my project plan while away from my computer.
- **NFR7 - C - Accessibility Compliance** As a student with disabilities, the system could meet WCAG 2.1 Level AA accessibility standards with keyboard navigation, screen reader support, and sufficient color contrast so that I can use the tool effectively regardless of my accessibility needs.
- **NFR8 - C - Scalability for Multiple Concurrent Users** As a student, the system could support at least 200 concurrent users without performance degradation so that I can access my Gantt chart reliably even during peak usage periods like the start of semester or project deadlines.

3.3 Evaluation Method

4 Feasibility

4.1 Stakeholder Meetings

4.2 Tools

4.3 Interface Mock-ups

[16]

5 Conclusion

conclude

References

[1] Academiac. 2018. *Are Gantt charts useful for PhD students?* https://academiac.net/2018/10/23/are-gantt-charts-useful-for-phd-students/ Published on academiac.net, Accessed: 14/11/2025.

[2] Asana. 2025. What is a Gantt Chart? Guide to Project Timelines. https://asana.com/resources/gantt-chart-basics. Accessed: 12/11/2025.

[3] Andrea Bednjanec and Martina Filipovic Tretinjak. 2013. Application of Gantt charts in the educational process. In *2013 36th International Convention on Information and Communication Technology, Electronics and Microelectronics (MIPRO)*. MIPRO, 631–635.

[4] Claude Besner and Brian Hobbs. 2008. Project management practice, generic or contextual: A reality check. *Project Management Journal* 39, 1 (2008), 16–33. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1002/pmj.20033 doi:10.1002/pmj.20033

[5] Mike Cohn. 2009. *User Stories Applied: For Agile Software Development.* Addison-Wesley, Boston.

[6] Methodist Collage. [n. d.]. UNIT-I: Interaction Paradigms: Computing Environments, Analysing Interaction Paradigms, Interaction Frameworks and Styles. https://methodist.edu.in. Human-Computer Interaction Course Materials, Methodist College. Accessed: 17/11/2025.

[7] Academy for International Modern Studies (AIMS). [n. d.]. What is Gantt Chart Project Management? https://aims.education/what-is-gantt-chart-project-management/. Accessed: 08/11/2025.

[8] GeeksforGeeks. 2025. *D3.js in JavaScript.* https://www.geeksforgeeks.org/javascript/d3-js/ Accessed: 25/09/2025.

[9] GeeksforGeeks. 2025. What is Gantt Chart in Project Management? https://www.geeksforgeeks.org/business-studies/what-is-gantt-chart-in-project-management/. Last updated: 23 Jul 2025. Accessed: 08/11/2025.

[10] Hela Hassen. 2024. The use of a project management tool in distance education to enhance students' engagement in group work. *Journal of Learning Development in Higher Education* 30 (Mar. 2024). doi:10.47408/jldhe.vi30.1146

[11] Innovare. 2023. Interactive Data Visualization in Education: A Complete Guide. https://innovaresip.com/resources/blog/interactive-data-visualization-in-education-guide/. Accessed: 17/11/2025.

[12] Charles Johnston and David Wierschem. 2007. Project Management Practices in the Information Technology Departments of Various Size Institutions of Higher Education. *Journal of International Technology and Information Management* 16 (01 2007). doi:10.58729/1941-6679.1209

[13] Justinmind. 2025. Principles and examples to master data visualization. https://www.justinmind.com/blog/data-visualization-examples-principles/. Accessed: 17/11/2025.

[14] Peter Landau. 2024. Gantt Chart vs. PERT Chart vs. Network Diagram: What Are the Differences? https://www.projectmanager.com/blog/gantt-vs-pert-vs-network-diagram. Published on projectmanager.com, Accessed: 14/11/2025.

[15] Page Laubheimer. 2020. Drag-and-Drop: How to Design for Ease of Use. https://www.nngroup.com/articles/drag-drop/. Nielsen Norman Group. Accessed: 18/11/2025.

[16] Visualising Data Ltd. 2025. Data Visualisation Resources. https://visualisingdata.com/resources/. Accessed: 25/09/2025.

[17] Kirti Mahajan and Leena Gokhale. 2018. Comparative Study of Static and Interactive VisualizationApproaches. *International Journal on Computer Science and Engineering* 10 (03 2018), 85–91. doi:10.21817/ijcse/2018/v10i3/181003016

[18] Eddie Meardon. 2025. *Gantt chart: Key features and benefits.* https://www.atlassian.com/agile/project-management/gantt-chart Published on atlassian.com, Accessed: 14/11/2025.

[19] Adrian Neumeyer. [n. d.]. Finish-to-Start Dependency Explained (with Examples). https://www.tacticalprojectmanager.com/finish-to-start-dependency/. Accessed: 12/11/2025.

[20] ProductPlan. 2020. What is MoSCoW Prioritization Method? Definition, Overview, and Best Practices. YouTube video. https://www.youtube.com/watch?v=pm4GbSRMElc&t=1s Accessed: 18/11/2025.

[21] ProjectManager. 2021. Gantt Chart Dependencies: Understanding Task Dependency Types. https://www.projectmanager.com/blog/gantt-chart-dependencies. Accessed: 12/11/2025.

[22] Diana Ramos. 2021. PERT Charts vs. Gantt Charts. https://www.smartsheet.com/content/pert-vs-gantt. Published on smarsheet.com, Accessed: 14/11/2025.

[23] Lulu Richter. 2025. Gantt Chart vs. Kanban Board. https://www.smartsheet.com/content/gantt-vs-kanban. Published on smarsheet.com, Accessed: 14/11/2025.

[24] Hélène Riol and Denis Thuillier. 2015. Project management for academic research projects: Balancing structure and flexibility. *International Journal of Project Organisation and Management* 7 (01 2015), 251. doi:10.1504/IJPOM.2015.070792

[25] Samyukta Sherugar and Raluca Budiu. 2024. Direct Manipulation: Definition. https://www.nngroup.com/articles/direct-manipulation/. Nielsen Norman Group. Accessed: 17/11/2025.

[26] Katie Shives. 2015. *Using Project Management Approaches to Tame Your Dissertation.* https://www.insidehighered.com/blogs/gradhacker/using-project-management-approaches-tame-your-dissertation Accessed: 08/11/2025.

[27] Smartsheet. 2020. *Gantt Chart with Dependencies.* https://www.youtube.com/watch?v=U4QftGiN5Nk Uploaded on YouTube by Smartsheet. Accessed: 30/09/2025.

[28] Coursera Staff. 2025. Gantt Charts: What They Are and How to Make Them. https://www.coursera.org/articles/gantt-charts. Accessed: 08/11/2025.

[29] TeamHub. 2023. *Discover the Benefits of an Interactive Gantt Chart Platform.* https://teamhub.com/blog/discover-the-benefits-of-an-interactive-gantt-chart-platform/ Published on TeamHub, Accessed: 14/11/2025.

[30] W3Schools. 2025. *D3.js Tutorial.* https://www.w3schools.com/js/js_graphics_d3js.asp Accessed: 25/09/2025.

[31] Kanika Wadhwa. 2024. *The role of Gantt chart in project management.* Master's thesis. Vaasa University of Applied Sciences.

[32] Tracey L. Weissgerber, Vesna D. Garovic, Marko Savic, Stacey J. Winham, and Natasa M. Milic. 2016. From Static to Interactive: Transforming Data Visualization to Improve Transparency. *PLOS Biology* 14, 6 (06 2016), 1–8. doi:10.1371/journal.pbio.1002484

[33] James M. Wilson. 2003. Gantt charts: A centenary appreciation. *European Journal of Operational Research* 149, 2 (2003), 430–437. doi:10.1016/S0377-2217(02)00769-5 Sequencing and Scheduling.

A Project Management

A.1 Gantt Chart

A.2 Risks

A.3 PLES Issues

B Generative AI